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TREND-BASED MOCK PAPER

GATE Ecology and Evolution

Mock Paper — 02

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Ecology and Evolution
Paper:	Mock 02
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Antonym of 'indigenous':

- (A) Native
- (B) Foreign
- (C) Local
- (D) Original

Section B: Ecology and Evolution

Q2. [1-MCQ] [1 mark]

Standard deviation is:

- (A) Mean
- (B) Spread
- (C) Median
- (D) Mode

Q3. [1-MCQ] [1 mark]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q4. [1-MCQ] [1 mark]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16

(D) 12

Q5. [1-MCQ] [1 mark]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q6. [1-MCQ] [1 mark]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q7. [1-MCQ] [1 mark]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q8. [1-NAT] [1 mark]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q9. [1-NAT] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

(Numerical Answer Type — Enter the numerical value)

Q10. [1-MCQ] [1 mark]

The vector $(1, 2, 3)$ projected onto $(2, 1, 0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q11. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q12. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q13. [2-NAT] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \sqrt{\langle u, u \rangle \langle v, v \rangle}$:

(Numerical Answer Type — Enter the numerical value)

Q14. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
 - (B) 2
 - (C) 16
 - (D) 1
-

Q15. [2-NAT] [2 marks]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

(Numerical Answer Type — Enter the numerical value)

Q16. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
 - (B) $y = x \ln(x)$
 - (C) $y = x \sin(\ln x)$
 - (D) $y = x^2$
-

Q17. [2-MCQ] [2 marks]

The Laplace transform of $t e^{-2t}$ is:

- (A) $1/(s+2)^2$
 - (B) $1/(s-2)^2$
 - (C) $s/(s+2)^2$
 - (D) $2/(s+2)^2$
-

Q18. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0, 1)$ is approximately:

- (A) 0.68
 - (B) 0.32
 - (C) 0.95
 - (D) 0.16
-

Q19. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
 - (B) 2
 - (C) 3
 - (D) 0
-

Q20. [2-MCQ] [2 marks]

The differential equation $y dx - x dy = 0$ has solution:

- (A) $y = Cx$
- (B) $y = C/x$
- (C) $y = Cx^2$
- (D) $x^2 + y^2 = C$

Q21. [2-NAT] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

(Numerical Answer Type — Enter the numerical value)

Q22. [2-NAT] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

(Numerical Answer Type — Enter the numerical value)

Q23. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot [1, 0]^T$ equals:

- (A) $[1, 0]^T$
 - (B) $[2, 1]^T$
 - (C) $[(3+2\sqrt{2})^{50}, \dots]$
 - (D) Cannot determine
-

Q24. [2-NAT] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

(Numerical Answer Type — Enter the numerical value)

Q25. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\operatorname{trace}(A^2)$ equals:

- (A) 14
 - (B) 6
 - (C) 36
 - (D) 12
-

Q26. [2-NAT] [2 marks]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
 - (B) 0.092
 - (C) 0.762
 - (D) 0.195
-

Q28. [2-MCQ] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
 - (B) $1/(s-2)^2$
 - (C) $s/(s+2)^2$
 - (D) $2/(s+2)^2$
-

Q29. [2-NAT] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

(Numerical Answer Type — Enter the numerical value)

Q30. [2-NAT] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

(Numerical Answer Type — Enter the numerical value)

Q31. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q32. [2-NAT] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \sqrt{\langle u, u \rangle \langle v, v \rangle}$:

(Numerical Answer Type — Enter the numerical value)

Q33. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q34. [1-MCQ] [1 mark]

Mendelian dihybrid cross (both heterozygous) ratio:

- (A) 3:1
- (B) 9:3:3:1
- (C) 1:2:1
- (D) 1:1:1:1

Q35. [1-MCQ] [1 mark]

Photosynthesis light reactions produce:

- (A) Glucose only
- (B) ATP and NADPH
- (C) CO₂
- (D) O₂ only

Q36. [1-MCQ] [1 mark]

Central dogma: reverse transcription is:

- (A) RNA $\rightarrow$ DNA
- (B) DNA $\rightarrow$ RNA
- (C) Protein $\rightarrow$ DNA
- (D) RNA $\rightarrow$ Protein

Q37. [1-MCQ] [1 mark]

Competitive enzyme inhibition: K_m

- (A) Increases
- (B) Decreases
- (C) Unchanged
- (D) Zero

Q38. [1-NAT] [1 mark]

DNA replication in eukaryotes occurs in:

(Numerical Answer Type — Enter the numerical value)

Q39. [2-MCQ] [2 marks]

Photosynthesis light reactions produce:

- (A) Glucose only
 - (B) ATP and NADPH
 - (C) CO₂
 - (D) O₂ only
-

Q40. [2-MCQ] [2 marks]

Central dogma: reverse transcription is:

- (A) RNA->DNA
 - (B) DNA->RNA
 - (C) Protein->DNA
 - (D) RNA->Protein
-

Q41. [2-NAT] [2 marks]

DNA replication in eukaryotes occurs in:

(Numerical Answer Type — Enter the numerical value)

Q42. [2-MCQ] [2 marks]

DNA replication in eukaryotes occurs in:

- (A) Cytoplasm
 - (B) Nucleus
 - (C) Mitochondria only
 - (D) G₁ phase
-

Q43. [2-NAT] [2 marks]

Central dogma: reverse transcription is:

(Numerical Answer Type — Enter the numerical value)

Q44. [2-MCQ] [2 marks]

Photosynthesis light reactions produce:

- (A) Glucose only
 - (B) ATP and NADPH
 - (C) CO₂
 - (D) O₂ only
-

Q45. [1-MCQ] [1 mark]

Hardy-Weinberg needs:

- (A) No mutation
 - (B) No migration
 - (C) Large pop
 - (D) All
-

Q46. [1-NAT] [1 mark]

mRNA processing:

(Numerical Answer Type — Enter the numerical value)

Q47. [1-MCQ] [1 mark]

Linked genes show:

- (A) Independent assortment
- (B) No recombination
- (C) Less than 50% recombination
- (D) Always 50%

Q48. [1-MCQ] [1 mark]

AaBb x aabb. Genotypic ratio:

- (A) 1:1:1:1
- (B) 3:1
- (C) 9:3:3:1
- (D) 1:2:1

Q49. [1-MCQ] [1 mark]

DNA B-form base pairs per turn:

- (A) 10
- (B) 10.4
- (C) 12
- (D) 8

Q50. [2-MCQ] [2 marks]

ABO blood group has how many alleles?

- (A) 2
- (B) 3
- (C) 4
- (D) 6

Q51. [2-NAT] [2 marks]

AaBb x aabb. Genotypic ratio:

(Numerical Answer Type — Enter the numerical value)

Q52. [2-MCQ] [2 marks]

DNA B-form base pairs per turn:

- (A) 10
- (B) 10.4
- (C) 12
- (D) 8

Q53. [2-MSQ] [2 marks]

AaBb x aabb. Genotypic ratio:

- (A) 1:1:1:1
- (B) 3:1
- (C) 9:3:3:1
- (D) 1:2:1

Q54. [2-NAT] [2 marks]

mRNA processing:

(Numerical Answer Type — Enter the numerical value)

Q55. [1-MCQ] [1 mark]

Logistic growth has:

- (A) No limit
- (B) Carrying capacity
- (C) Exponential
- (D) None

Q56. [1-MCQ] [1 mark]

Trophic level energy transfer efficiency:

- (A) 10%
- (B) 50%
- (C) 90%
- (D) 1%

Q57. [1-MCQ] [1 mark]

Logistic growth equation has carrying capacity:

- (A) $dN/dt = rN$
- (B) $dN/dt = rN(1-N/K)$
- (C) $dN/dt = r$
- (D) $dN/dt = K-N$

Q58. [1-MCQ] [1 mark]

Most abundant greenhouse gas:

- (A) CO₂
- (B) CH₄
- (C) N₂O
- (D) Water vapor

Q59. [2-NAT] [2 marks]

Most abundant greenhouse gas:

(Numerical Answer Type — Enter the numerical value)

Q60. [1-MCQ] [1 mark]

Darwin finches from:

- (A) Galapagos
- (B) Africa
- (C) Asia
- (D) Australia

Q61. [1-MCQ] [1 mark]

Biodiversity hotspot criteria:

- (A) Endemism
- (B) Deforestation
- (C) Both
- (D) None

Q62. [1-NAT] [1 mark]

Mitochondria function:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-MCQ] [1 mark]

Photosynthesis in:

- (A) Mitochondria
- (B) Chloroplast
- (C) Nucleus
- (D) Ribosome

Q64. [1-MCQ] [1 mark]

Binomial nomenclature by:

- (A) Linnaeus
- (B) Darwin
- (C) Mendel
- (D) Watson

Q65. [1-NAT] [1 mark]

Monarch butterfly migrates from:

(Numerical Answer Type — Enter the numerical value)

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1MCQ	B	1
3	1MCQ	B	1
4	1MCQ	A	1
5	1MCQ	D	1
6	1MCQ	C	1
7	1NAT	B	1
8	1MCQ	B	1
9	1MCQ	A	1
10	1NAT	A	1
11	1MCQ	A	1
12	1MCQ	B	1
13	1MCQ	B	1

14	1MCQ	D	1
15	1MCQ	B	1
16	1MCQ	A	1
17	1MCQ	C	1
18	1MCQ	C	1
19	1MCQ	C	1
20	1NAT	B	1
21	1MCQ	A	1
22	1MCQ	C	1
23	1MCQ	A	1
24	1MCQ	B	1
25	1MCQ	A	1
26	1MCQ	A	1
27	1NAT	B	1
28	1NAT	A	1

29	1NAT	A	1
30	1MCQ	B	1
31	2MCQ	C	2
32	2MCQ	B	2
33	2MCQ	B	2
34	2NAT	A	2
35	2MCQ	A	2
36	2NAT	B	2
37	2MCQ	A	2
38	2MCQ	A	2
39	2MCQ	A	2
40	2MCQ	B	2
41	2MCQ	B	2
42	2NAT	B	2
43	2NAT	A	2

44	2MCQ	A	2
45	2NAT	C	2
46	2MCQ	A	2
47	2NAT	A	2
48	2MCQ	A	2
49	2MCQ	B	2
50	2NAT	D	2
51	2MCQ	A	2
52	2NAT	B	2
53	2MCQ	A	2
54	2NAT	A	2
55	2MCQ	B	2
56	2NAT	A	2
57	2NAT	A	2
58	2MCQ	A	2

59	2MSQ	A	2
60	2MCQ	B	2
61	2NAT	A	2
62	2NAT	A	2
63	2MCQ	C	2
64	2NAT	B	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Statistics | Difficulty: moderate

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Ecology | Topic: Population | Difficulty: easy

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: Evolution | Topic: Natural Selection | Difficulty: difficult

Q5.

Answer: D

Solution:

Correct Answer: D

Subject: Genetics | Topic: Population Genetics | Difficulty: easy

Q6.

Answer: C

Solution:

Correct Answer: C

Subject: Environmental Science | Topic: Biodiversity | Difficulty: moderate

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Cell Biology | Topic: Mitochondria | Difficulty: moderate

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Plant Physiology | Topic: Photosynthesis | Difficulty: easy

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Zoology | Topic: Taxonomy | Difficulty: easy

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Behavioral Ecology | Topic: Migration | Difficulty: difficult

Q11.

Answer: A

Solution:

10% law: ~10% energy transferred between trophic levels (Lindeman).

Subject: Ecology | Topic: Ecology | Difficulty: easy

Q12.

Answer: B

Solution:

AaBb x AaBb: phenotypic ratio 9:3:3:1.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q13.

Answer: B

Solution:

Logistic: $dN/dt = rN(1-N/K)$. K is carrying capacity.

Subject: Ecology | Topic: Ecology | Difficulty: moderate

Q14.

Answer: D

Solution:

Water vapor: most abundant GHG by concentration and radiative forcing.

Subject: Ecology | Topic: Ecology | Difficulty: moderate

Q15.

Answer: B

Solution:

Light reactions: ATP + NADPH for Calvin cycle. O₂ is byproduct.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q16.

Answer: A

Solution:

Reverse transcription: RNA→DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: easy

Q17.

Answer: C

Solution:

Rolle's: f continuous on [a,b], differentiable on (a,b), f(a)=f(b). Then f'(c)=0 for some c in (a,b).

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q18.

Answer: C

Solution:

Kirchhoff: for K_n, spanning trees = $n^{(n-2)} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q19.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q20.

Answer: B

Solution:

Pre-mRNA: 5 cap, intron splicing, 3 poly-A tail.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q21.

Answer: A

Solution:

Competitive: V_{max} unchanged, K_m increases (more substrate needed).

Subject: Biology | Topic: Biology | Difficulty: easy

Q22.

Answer: C

Solution:

Linked genes on same chromosome: recombination < 50%.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q23.

Answer: A

Solution:

AaBb x aabb: offspring AaBb, Aabb, aaBb, aabb in 1:1:1:1.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q24.

Answer: B

Solution:

B-DNA: ~10.4 bp per turn.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q25.

Answer: A

Solution:

$L\{e^{-(s+a)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q27.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q28.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q29.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q30.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) \cdot w = (4/5) \cdot (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q32.

Answer: B

Solution:

$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q33.

Answer: B

Solution:

ABO: I^A , I^B , i (three alleles).

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q34.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q35.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q36.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q38.

Answer: A

Solution:

$L\{t^2 f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q41.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: B

Solution:

$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q43.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q45.

Answer: C

Solution:

$y = A\cos(2x) + B\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q46.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q48.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q49.

Answer: B

Solution:

Light reactions: ATP + NADPH for Calvin cycle. O_2 is byproduct.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q50.

Answer: D

Solution:

Water vapor: most abundant GHG by concentration and radiative forcing.

Subject: Ecology | Topic: Ecology | Difficulty: easy

Q51.

Answer: A

Solution:

Reverse transcription: RNA $\rightarrow$ DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: easy

Q52.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: moderate

Q53.

Answer: A

Solution:

$L\{f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q54.

Answer: A

Solution:

$AaBb \times aabb$: offspring $AaBb$, $Aabb$, $aaBb$, $aabb$ in 1:1:1:1.

Subject: Genetics | Topic: Genetics | Difficulty: difficult

Q55.

Answer: B

Solution:

B-DNA: ~10.4 bp per turn.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q56.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q57.

Answer: A

Solution:

$L\{t^k f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q58.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=xv$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q59.

Answer: A

Solution:

AaBb x aabb: offspring AaBb, Aabb, aaBb, aabb in 1:1:1:1.

Subject: Genetics | Topic: Genetics | Difficulty: easy

Q60.

Answer: B

Solution:

DNA replication: S phase in nucleus. Mitochondrial DNA replicates separately.

Subject: Biology | Topic: Biology | Difficulty: easy

Q61.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q62.

Answer: A

Solution:

Reverse transcription: RNA $\rightarrow$ DNA (retroviruses like HIV).

Subject: Biology | Topic: Biology | Difficulty: moderate

Q63.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q64.

Answer: B

Solution:

Pre-mRNA: 5 cap, intron splicing, 3 poly-A tail.

Subject: Genetics | Topic: Genetics | Difficulty: moderate

Q65.

Answer: B

Solution:

Light reactions: ATP + NADPH for Calvin cycle. O₂ is byproduct.

Subject: Biology | Topic: Biology | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 32 questions, 55 marks
Biology: 11 questions, 17 marks
Genetics: 10 questions, 15 marks
Ecology: 5 questions, 6 marks
General Aptitude: 1 questions, 1 marks
Evolution: 1 questions, 1 marks
Environmental Science: 1 questions, 1 marks
Cell Biology: 1 questions, 1 marks
Plant Physiology: 1 questions, 1 marks
Zoology: 1 questions, 1 marks
Behavioral Ecology: 1 questions, 1 marks

Difficulty Distribution:

Easy: 19 | Moderate: 39 | Difficult: 7

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.